

4(a)	any three from: max 2 for similarities, max 2 for differences similarities, both have phospholipids ; channel / transport, protein ; A intrinsic / integral / transmembrane, protein R carrier / pump, protein a bilayer / two layers ; differences do not need to have 'outer membrane' 'bacterial' if context is clear (bacterial outer membrane) has lipopolysaccharides ; ora (cell surface membrane / eukaryote) has, phospholipid bilayer / both layers with phospholipids ; ora if ora also given (bacterial) only has outer layer of (outer) membrane with phospholipids, can also award mp1 AVP ;; e.g. cell surface membrane has, glycocalyx / glycolipids / glycoproteins cell surface membrane has cholesterol outer membrane does not have extrinsic proteins	3
4(b)(i)	any one from: formation of O-IMV involves a more complex process ; e.g. greater variety of, contents / cell structures R organelles ref. to needing to make a vesicle with two 'membranes' / AW A 'layers' peptidoglycan, chains / layer / bonds, need to be broken ; A peptidoglycan wall needs to be broken causes, (temporary) weakness in cell wall / risk of lysis ; idea that more resources required to form vesicle or too great a loss of important cell, components / structures ref. to take more time to produce ; uses (more) energy to, break wall / repair wall / make the vesicle ; I ref. to organelles AVP ; e.g. suggestion that periplasmic proteins need to be released in greater numbers so more OMVs formed	1
4(b)(ii)	any one suggestion that: must be in context of a fully formed O-IMV function of O-IMVs require, energy / ATP ; A suggested function e.g. DNA transcription / metabolic reactions / for active transport I cell processes O-IMV (formation) has cytoplasm, which contains ATP ; OMVs formed only from periplasm, which has no ATP ; A no ATP in outer membrane space AVP ; e.g. no / few, ATP transporters / transport proteins, for exit of ATP into periplasm (in area where OMV forms)	1

Question	Answer	Marks
4(b)(iii)	<i>ref. to</i> DNA in O-IMVs being passed on to other bacteria ; e.g. O-IMVs may, fuse with / be taken up by / AW, plasmids / DNA, in (degraded) O-IMVs may enter other bacteria qualified <i>ref. to</i> O-IMV DNA and antibiotic resistance ; e.g. O-IMVs have, alleles / genes / plasmid, that, give / confer / AW, antibiotic resistance DNA may mutate to give (a, gene / allele, for) antibiotic resistance DNA, codes for / expresses / AW, a protein involved in antibiotic resistance example of gene conferring resistance e.g. enzyme to breakdown antibiotic change to ribosome structure (to prevent antibiotic attachment) efflux pump for / membrane protein to pump out, antibiotic	2